EE115 Analog Circuits
模拟电路

Dr. Hao Ren 任豪

Office: SIST Bldg 3-328

renhao@shanghaitech.edu.cn

Outline

- Course Organization and Calendar
- Textbooks and supplementary materials
- Need of analog electronics
- Why is it more challenging to design analog than digital circuitry?
- Why integrated and why CMOS?

Course Organization and Calendar

- ALL UPDATES on EE115 Blackboard
- Instructors:
  - Dr. Hao Ren 任豪 Office: SIST Bldg 3-328
  - renhao@shanghaitech.edu.cn
- 10:15-11:55am, Thursdays（1~14周）， Tuesdays（2~12周的双周）
- 地点：1D-104
- Homework:
  - Posted in **Blackboard**
  - Due date is on the blackboard, usually one week after it is posted
  - Submission in Blackboard.
  - Late homework will be discounted by 20% per day
  - You may discuss homework problems with other students in the class, the TAs, or the instructor.
  - The work you submit for grading must be your own.

Grades

- Homework: 20 pts (individual)
- Midterm exam: 20 pts (individual)
- Final exam: 20 pts (individual)
- Research project: 20 pts(individual)
- Experiments: 20 pts (刘闯老师负责)
- Cheating will result in automatic Fail!

TA:

- 陈明东chenmd2025@shanghaitech.edu.cn
- 娄洋louyang2024@shanghaitech.edu.cn
- 肖海川xiaohch2025@shanghaitech.edu.cn
- 李明鑫limx12024@shanghaitech.edu.cn

Textbooks & Supplementary Readings

Textbooks

- *Microelectronic Circuits – Adel S Sedra & Kenneth Carless Smith*
- *Design of Analog CMOS Integrated Circuits – Razavi Behzad*

Supplementary Reading

- *Analysis and Design of Analog Integrated Circuits– Paul R. Gray, Paul J. Hurst, Stephen H. Lewis, Robert G. Meyer*

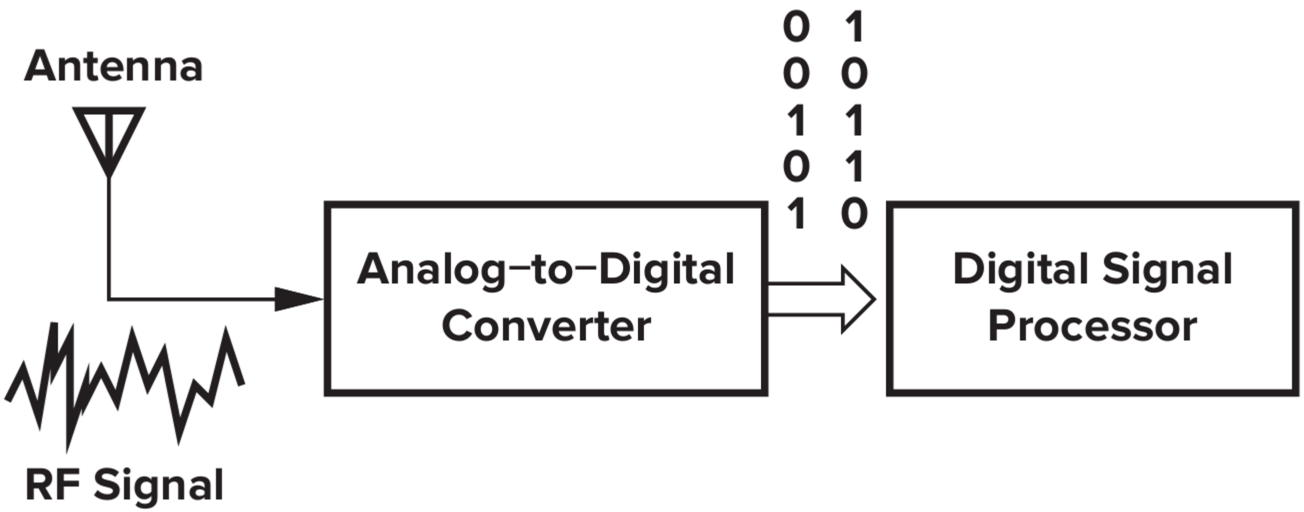
Need of analog electronics

Need of analog electronics

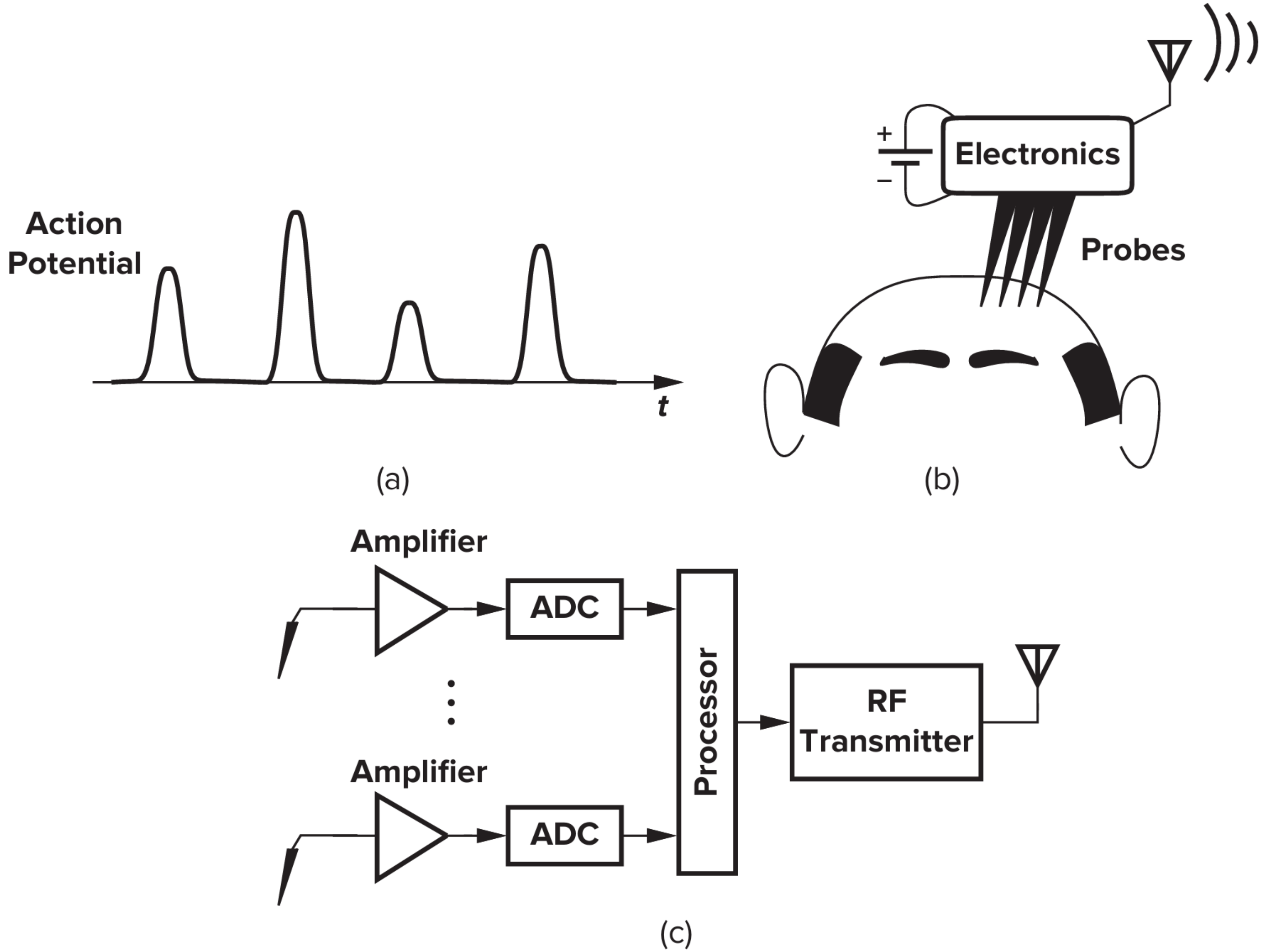
- Since early 1980s, many functions in analog domain have migrated to digital domain as DSP and IC fabrication techniques improve tremendously. Then why do we need "analog"?

Examples:

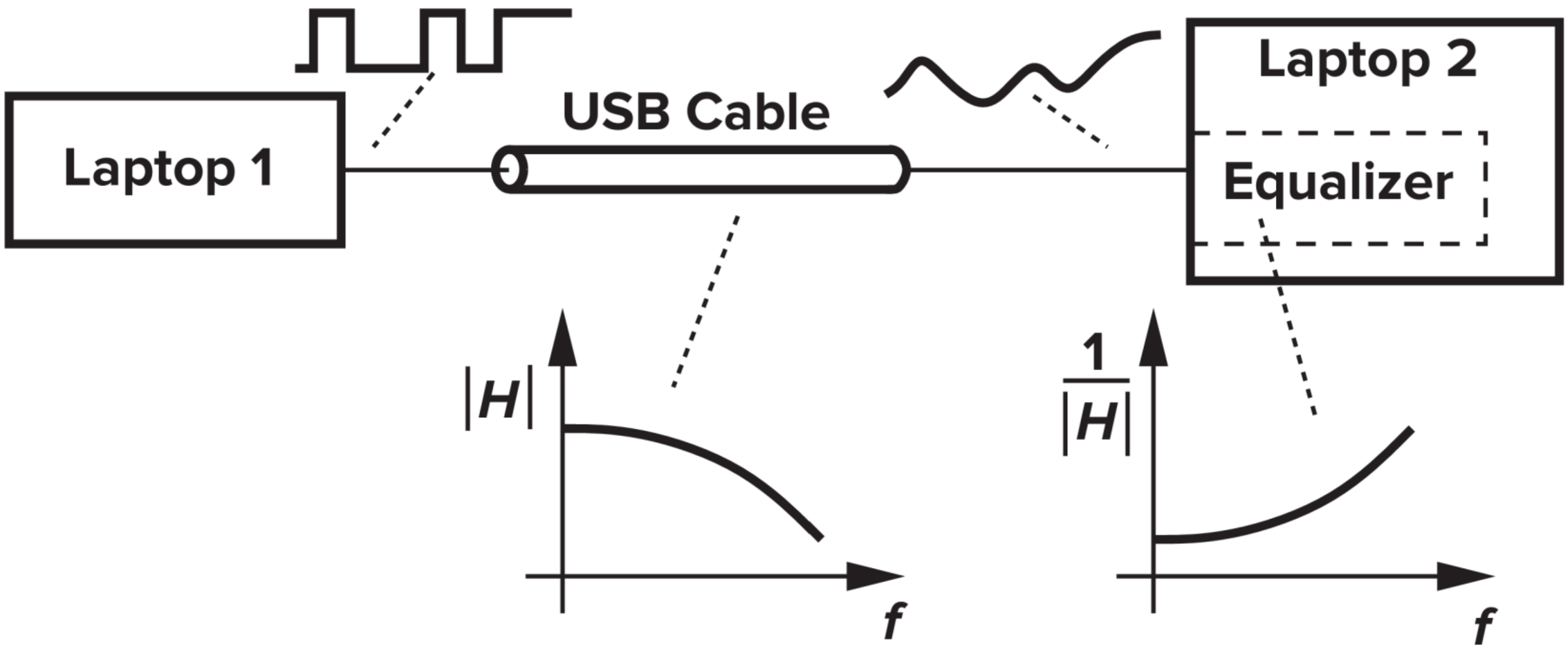
- Processing of natural signals such as microphone, photocells, seismographic sensor, etc.



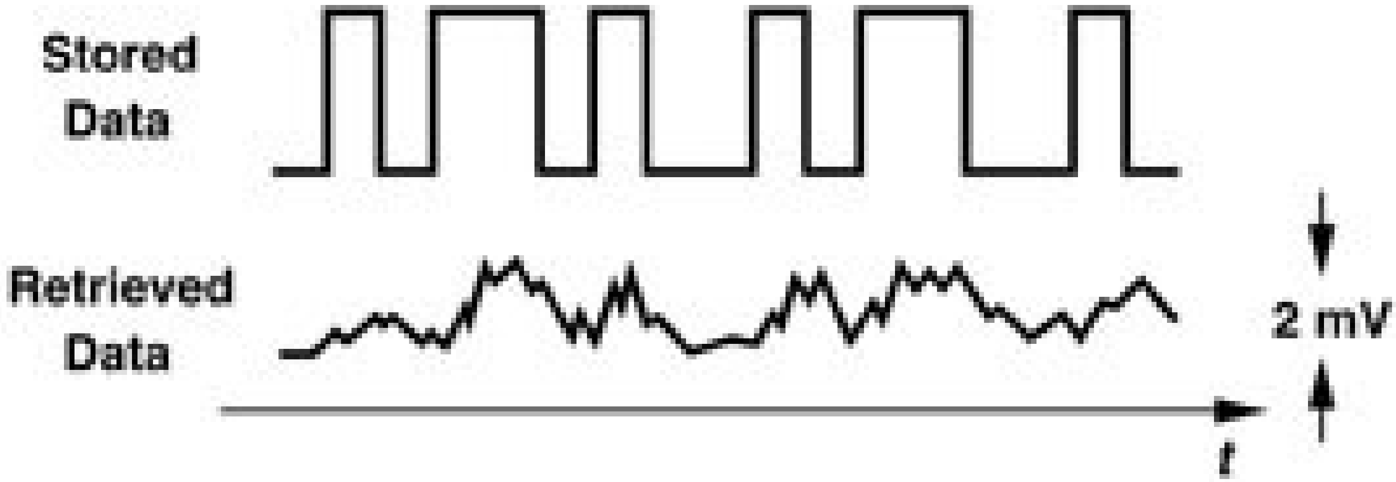
Need of analog electronics-neural recording



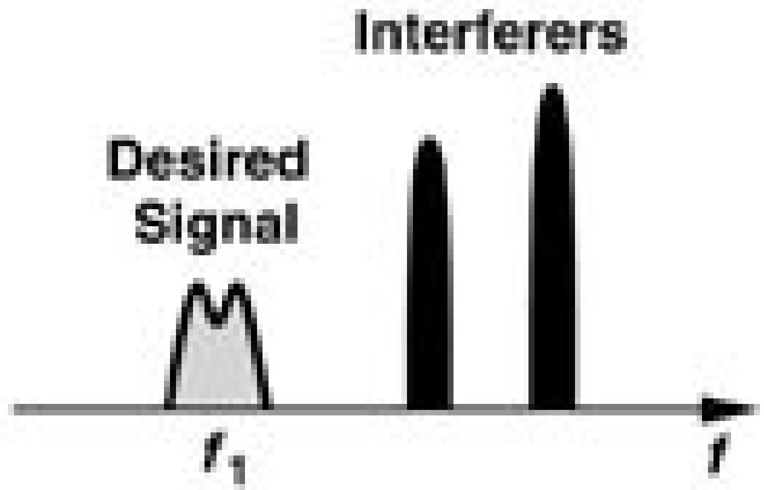
Digital communication



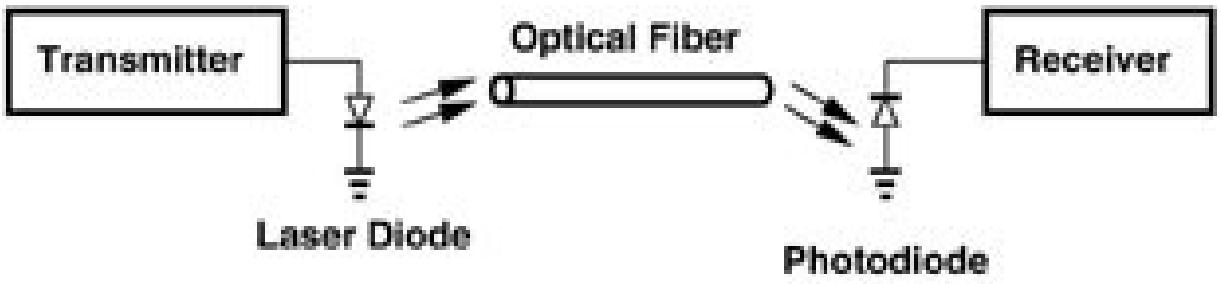
Disk drive electronics



Wireless receivers



Optical receivers



Why is it more challenging to design analog than digital circuitry?

Analog is more challenging than digital

- Digital entails primarily ONE trade-off: speed and power, analog face multiple trade-offs including speed, power, gain, precision, etc.
- Analog is significantly more sensitive to noise, crosstalk, and other interferences than digital.
- Second-order effects in devices influence more to analog than digital.
- Almost zero automation in analog, meaning every device needs to be hand-crafted.

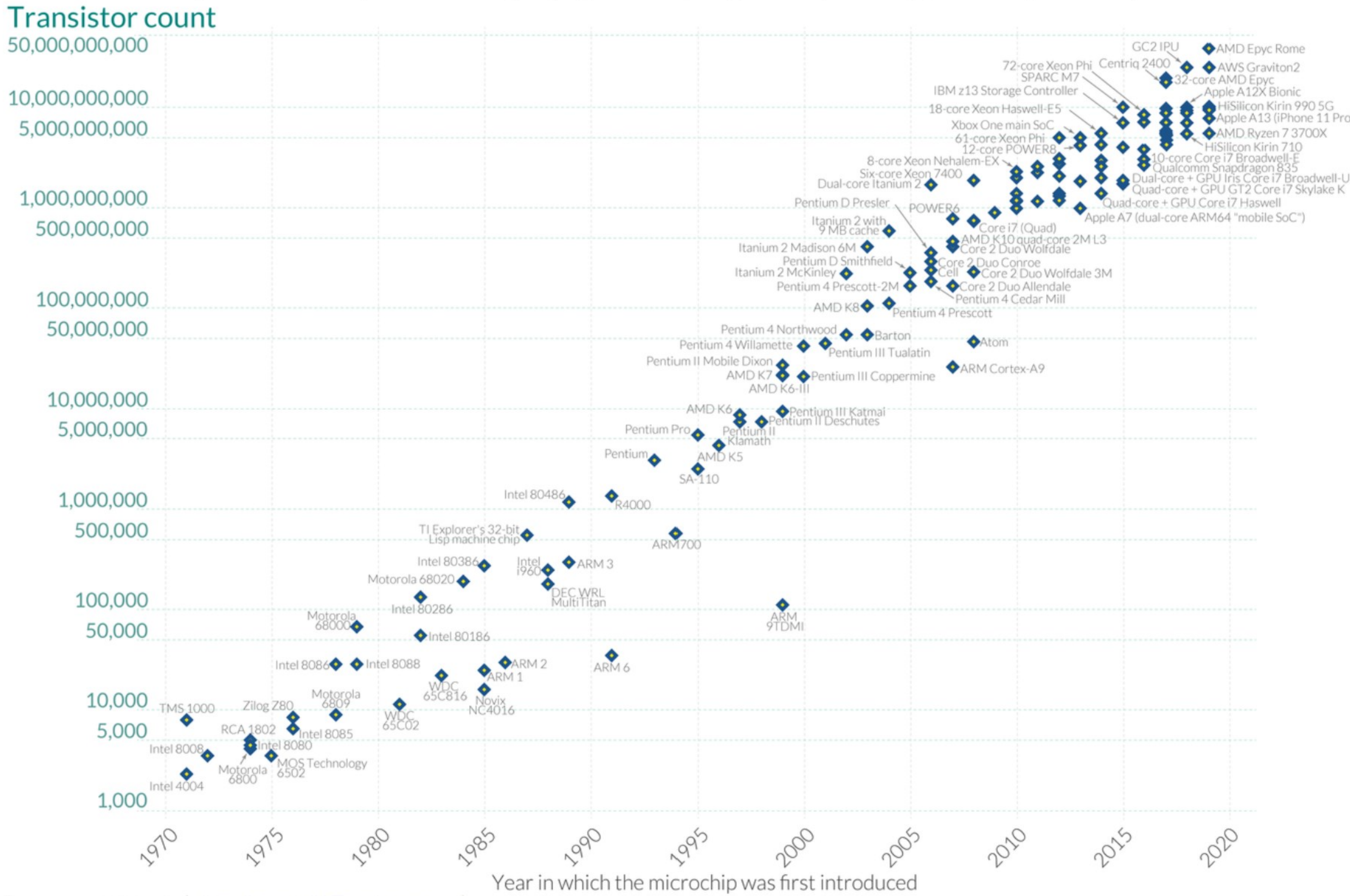
Why integrated and why CMOS?

Moore’s Law

Moore’s Law: The number of transistors on microchips has doubled every two years

Our World in Data

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

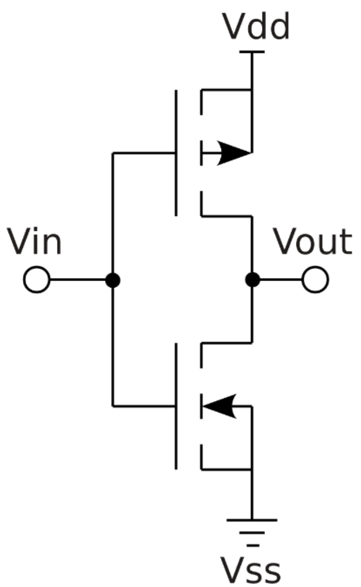
OurWorldinData.org – Research and data to make progress against the world’s largest problems.

Analog is more challenging than digital

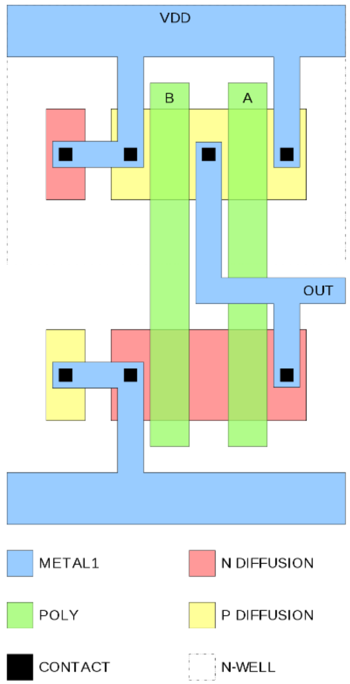
- Size of a transistor, gate length, has been scaled down incredibly, from 25 μm in 1960 to 2 nm in 2026.
- MOSFETs was patented in early 1930s far before the invention of bipolar transistors; technology becomes practical later in early 1960s. Yet, only n-type MOSFETs were produced. CMOS becomes available in mid-1960s
- CMOS dissipates power only during switching and requires very few devices in contrast to bipolar or GaAs.
- However, MOSFETs were slower and noisier than biopolar; i.e., gm of MOSFETs << gm of BJT
- Scalability and speed of CMOS has championed for past 30 years !

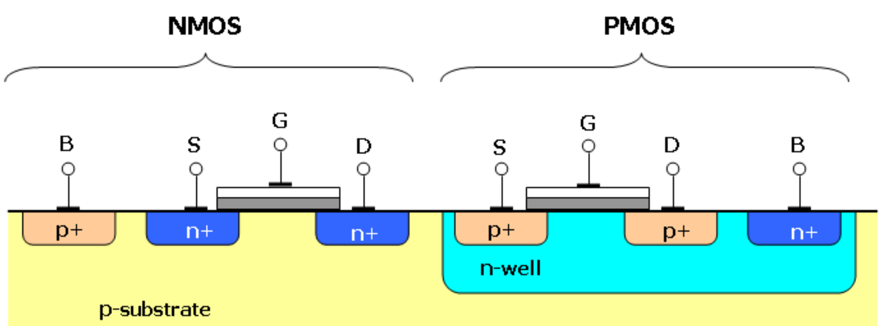
CMOS Circuit

Inverter

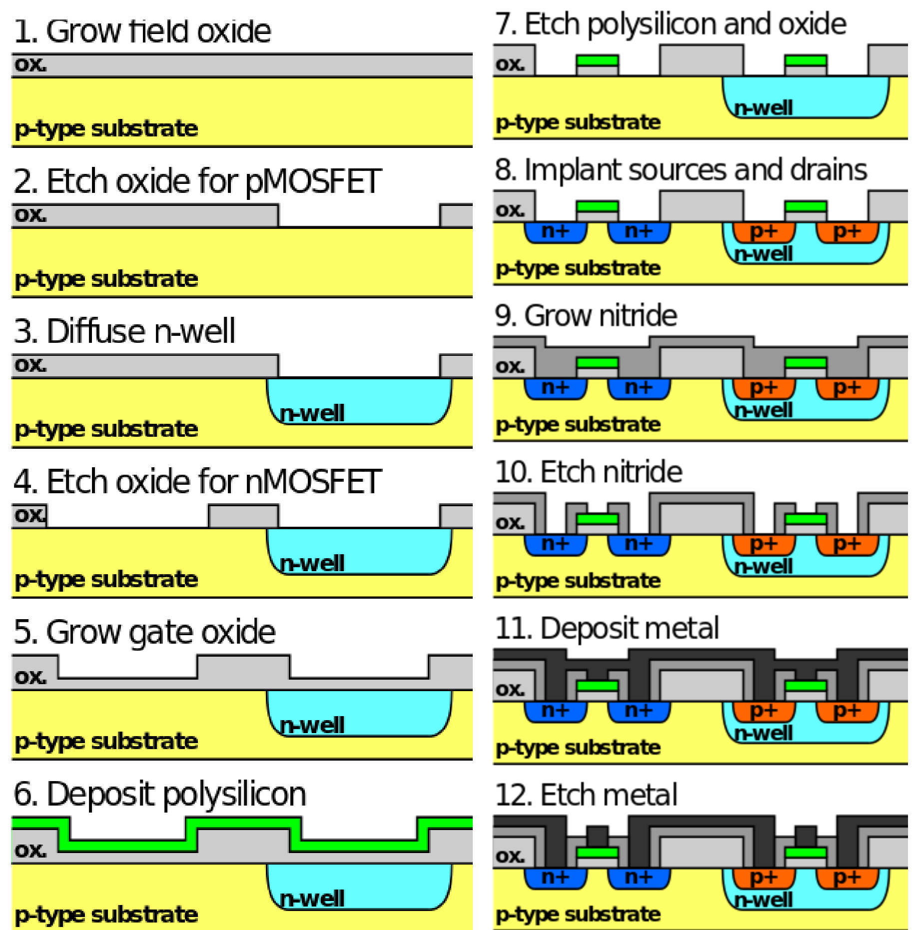


Layout





Process Flow



ShanghaiTech circuit courses

- Circuits
- Analog Circuits
- Digital Circuits
- Analog Integrated Circuits I
- Digital Integrated Circuits I
- Advanced Analog Integrated Circuits (graduate course)
- Advanced Digital Integrated Circuits (graduate course)